E (a) How the surrogate check works

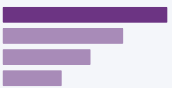
1. Start from each patient's out-of-bag risk score.



2. A random forest learns that score from clinical (21 BLCA, 25 BRCA) and first-order pathomic features (504, 514).



3. Rank features and tissue regions by importance (impurity).



4. Two checks: does the order survive a second estimator, and does each feature predict observed PFI after false-discovery-rate control?

permutation importance

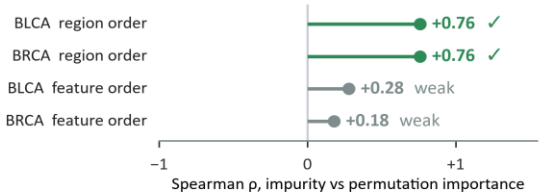
Cox vs PFI, FDR

WHY IT MATTERS

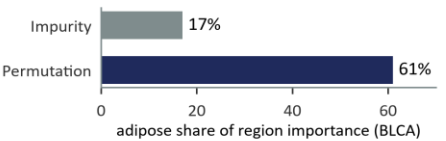
The surrogate explains the model's score. Only features that pass the outcome test say something about the disease.

(b) What the audit found

A second estimator: region order holds



Same leading region, but its share is not robust



Against outcome, no feature survives correction

